

Spatial Cluster Randomized Trials - Sampling Design with Spillover Effects & Spatial Dependence

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Abstract

Background: Investigators designing cluster randomized trials desire insights directing treatment assignment methodology for studies involving spillover effects and spatial dependence.

Methods & Simulation: Treatment assignment strategies including simple random sampling (SRS) and block stratified sampling (BSS) are defined and spatial autoregressive modeling is applied with consideration for spillover effects and spatial dependence for estimation of intervention effects. A simulation study is carried out comparing SRS and BSS sampling methods on spatial grids of varying sizes. A range of spillover effects and levels of spatial dependence were considered for estimation of the intervention effect via a spatial autoregressive (SAR) model.

Results: Findings of an extensive simulation study comparing simple random sampling and block stratification methods indicate that randomly selected treatment assignments result in best case reduced Mean Squared Error (MSE) when estimating intervention effects, but block stratified treatment assignments lead to minimal variation in MSE among a series of treatment combinations, indicating that a block stratified treatment arrangement won't achieve the minimal level of estimation error, but it remains robust across a range of selected parameters. This advantage of BSS

is conditional rather than universal: when spillover is bidirectional among treated units and background spatial dependence is present, the rigid spatial separation of BSS can instead amplify bias, and the bias under designs that render the spillover effect non-identifiable can be substantial (on the order of the spillover magnitude itself). Even though SRS is consistent and unbiased with reduced MSE, we consider variation among all possible treatment combinations to ensure a robust result.

Conclusion: The relationship between spillover effects and mean squared errors (MSE) of intervention effect estimation is apparent. The MSE for the intervention effect, which is the average MSE over each of N simulation iterations, is minimized for some combinations of random sampling treatment assignment, but block stratified assignment minimizes variance among combinations of possible treatment arrangements. In short, the SRS technique may achieve minimum average MSE in some cases, while BSS achieves respectable average estimation error with minimal variation between treatment combinations when spillover flows primarily into control clusters. The choice between the two designs is therefore conditional on the anticipated spillover mechanism, and where bias is severe, analysis-stage buffer-zone designs are a promising complementary strategy.

1 Introduction

1.1 Background & Motivation

Randomized controlled trials (RCTs) represent the gold standard for establishing causal inference when assessing the efficacy and safety of new interventions[1][2]. However, in many real-world scenarios, particularly in public health, education, and the social sciences, randomizing individuals is not feasible due to logistical constraints, ethical considerations, or the inherent nature of the intervention itself, which may be administered at a community level. In such cases, the cluster randomized trial (CRT) emerges as the preferred and most rigorous design, where intact social units or groups such as villages, schools, or clinics are randomized to intervention or control arms[3][4].

A key advantage of the CRT design is its ability to evaluate interventions where there is high risk of contamination between treated and untreated individuals within the same cluster[3][5]. Furthermore, CRTs are uniquely positioned to allow for the estimation of not only the direct effects of an intervention on those who receive it but also the indirect spillover effects. These are the additional impact of an intervention on individuals who did not directly receive it, typically including those in control clusters.

Measuring these spillover effects is often of primary scientific and policy interest, as it provides a comprehensive assessment of an intervention’s total public health impact[3][6]. When trial clusters are defined by geographic areas, as is frequently the case, the design must contend with inherent spatial complexities that can challenge the core statistical assumption of cluster independence. Two primary and interrelated spatial phenomena can profoundly influence trial outcomes and compromise the validity of the results: spillover effects and spatial dependence[3][2].

Spillover occurs when an intervention’s effects in one cluster extend or “spill over”, into neighboring clusters, creating significant analytical challenges[7][8]. This is especially common in trials of infectious disease interventions. A study on the use of permethrin-impregnated bed nets by Binka et al. (1998) provided early, compelling evidence of this phenomenon, demonstrating that child mortality was significantly lower in control villages situated near intervention villages, presumably due to a reduction in the local mosquito population[9][10][11]. Similarly, the work by Miguel and Kremer (2004) on a school-based de-worming program in Kenya found significant positive externalities, showing that the health and school attendance of untreated children improved if they lived near schools that were part of the intervention[12]. The critical implication is that failing to account for such positive spillover can lead to a substantial underestimation of the intervention’s true effect, as the measured outcomes in the control group are artificially improved by their proximity to the intervention arm. Compounding this issue is spatial dependence, a fundamental concept supported by Tobler’s First Law of Geography that asserts nearby observations tend to be more related than distant ones[8][13]. In the context of a CRT, this principle suggests that outcomes in nearby clusters may be correlated due to unmeasured, spatially-patterned environmental, social, or demographic factors. The presence of this spatial autocorrelation violates the assumption that clusters are independent units, which can lead to biased standard errors and an increased Type I error rate undermining statistical inference[14][15][13].

In response to these challenges, a body of literature has emerged focusing on post-hoc analytical adjustments applied after the data are collected. A systematic review by Jarvis et al. (2017) groups these into two categories[8]: the *spatial variable* approach, which constructs a distance- or density-based covariate—for example the distance to the nearest intervention unit[11][10], outcome rates across distance bands[9], the density of treated neighbors within a fixed radius[12][16], or a “potential exposure” metric[17]—and the more elaborate *spatial modeling* approach, which embeds the geographic structure

directly into the model through a distance-decay covariance[18] or spatial weights that treat correlation as an unobserved process[19]. A distinct framework instead randomly assigns intervention areas and models treatment as a distance-based continuous dose, estimating a dose-response function rather than a cluster-level intervention or spillover effect[20].

Despite the growing sophistication of these analytical tools, they are fundamentally post-hoc corrective solutions to a problem that could (and should) be addressed proactively at the design stage. The existing literature reveals a noticeable lack of research into how the initial sampling and allocation of treatments across a geographic landscape can be optimized to manage these known spatial complexities. The review by Jarvis et al. (2017) highlights this gap, noting that in practice, the impact of location on trial results is definitely an afterthought and oftentimes entirely ignored[8]. In fact, trial designs explicitly intended to mitigate spatial effects, such as creating well-separated clusters or buffer zones, were so uncommon that studies employing them were excluded from their primary analysis, underscoring how peripheral these considerations have been to standard practice[8].

This oversight has significant consequences because known studies that compare spatial analytical approaches consistently find that accounting for spatial effects is important. Silcocks and Kendrick (2010) demonstrated that spatial models not only fit primary care trial data better than standard CRT models, but also that their inclusion altered both the intervention effect point estimate and its precision[19]. Similarly, Chao et al. (2015) found that adjusting for their spatial exposure metric lowered the estimated effect of a typhoid vaccine[17]. The logical conclusion is that standard trial designs that rely on the simple random allocation of clusters, may be systematically producing biased or inefficient estimates without investigators being aware. This leads to the central question motivating the work of our study. That is, “How does the choice of an *a priori* treatment assignment sampling strategy impact the estimation of intervention effects in the presence of both spatial spillover effects and spatial dependence?” While we have methods to analyze the problem after a study is completed, there is currently a lack of evidence-based guidance for investigators on how to design a cluster randomized trial to mitigate these issues from the outset of study planning.

1.2 Aims & Contributions

This study aims to address this critical gap by moving beyond post-hoc analysis to investigate the important role of treatment sampling design in spatial CRTs. Through a comprehensive simulation study, we systematically compare the performance of two contrasting approaches for assigning study clusters to intervention and control arms: simple random sampling (SRS), representing the standard practice for treatment assignment, and a block stratified sampling (BSS) method. The BSS strategy is intentionally designed to create spatial separation between clusters of the same treatment assignment by isolating them to potentially reduce the confounding influence of spillover and spatial dependence.

Our simulations explore how the estimation error of the intervention effect, quantified by Mean Squared Error which is composed of bias and variance, varies across a range of realistic conditions including different magnitudes of spillover effects, the presence or absence of underlying spatial dependence, and variation of restrictions on the geographic distance over which spillover can occur. By identifying the systematic patterns that emerge, this novel research provides practical, evidence-based guidance for investigators designing future spatial CRTs. The findings will equip researchers with the knowledge to select sampling strategies that align with their specific study goals, ultimately leading to more robust, efficient, and valid estimates of intervention effects in complex spatial settings.

The remainder of this article is organized as follows: Section 2 outlines the theoretical framework of the study, including the data generation process, treatment sampling procedures, and estimation methods employed. Section 3 details the procedure and results of the simulation study undertaken to evaluate the sampling methods considered. Section 4 considers an application of these findings to make a recommendation for other investigators. Finally, Section 5 discusses the implications of our findings, potential drawbacks to the work performed, and potential areas for continued future research.

2 Design of Spatial CRTs

2.1 Cluster Randomized Trials

The randomized controlled trial (RCT) serves as the gold standard design to evaluate intervention efficacy[21]. Its core principle, the random allocation of individual participants, is expected to balance both measured and unmeasured confounding variables across treatment arms[22]. However, individual

randomization is often impractical, or an intervention is delivered at a group level, necessitating an alternative approach[4][3]. In these scenarios, the cluster randomized trial (CRT) is employed, where intact social units or groups like households, clinics, or geographic areas are randomized to intervention arms[4][23]. A key advantage of this design is the limitation of treatment contamination, where control group members are inadvertently exposed to the intervention, which would otherwise bias the estimated treatment effect toward the null[3]. Statistical considerations in a CRT are the measured response from an intervention and the inherent correlation among outcomes of individuals within the same cluster, a dependency quantified by the intraclass correlation coefficient (ICC)[4][24]. The ICC measures how similar the outcomes of individuals within a cluster are likely to be, relative to those of other clusters. This intra-cluster correlation violates the standard assumption of independence between observations. The failure to account for this feature in the analysis leads to underestimated standard errors and an inflated probability of a Type I error[25][26]. Therefore, the valid design and analysis of CRTs require specialized statistical methods that properly account for this clustering structure, with comprehensive frameworks provided by Donner & Klar (2000)[4] and Hayes & Moulton (2017)[3].

2.2 Neighbor Definitions

2.2.1 Contiguity-Based

Contiguity-based neighbors[27] are constructed with the assumption that neighbors of a given area (individual cells in the case of this study) are other clusters (or cells) that share a common boundary. This boundary may be an edge and/or vertex. The **spdep** R package[28][29][30][31] includes functions for working with spatial neighbors and spatial dependence structures. In this study, we consider Rook Neighbors (shown in Figure 1 (a)), which fulfill a contiguity condition where a pair of clusters share a common boundary made up by an edge. Queen neighbors (shown in Figure 1 (b)) are another type of contiguity-based neighbors that fulfill a contiguity condition where a pair of clusters share a common boundary of either an edge and/or a vertex[27]. Clusters may also have Bishop neighbors where a common boundary at the vertex (corner) of the spatial region is shared. Clusters may also have higher-order neighbors where they do not directly share an edge or vertex, but rather have a common boundary with a cluster that is a neighbor of another cluster. In other words, these are “neighbors of neighbors” of the reference cluster.

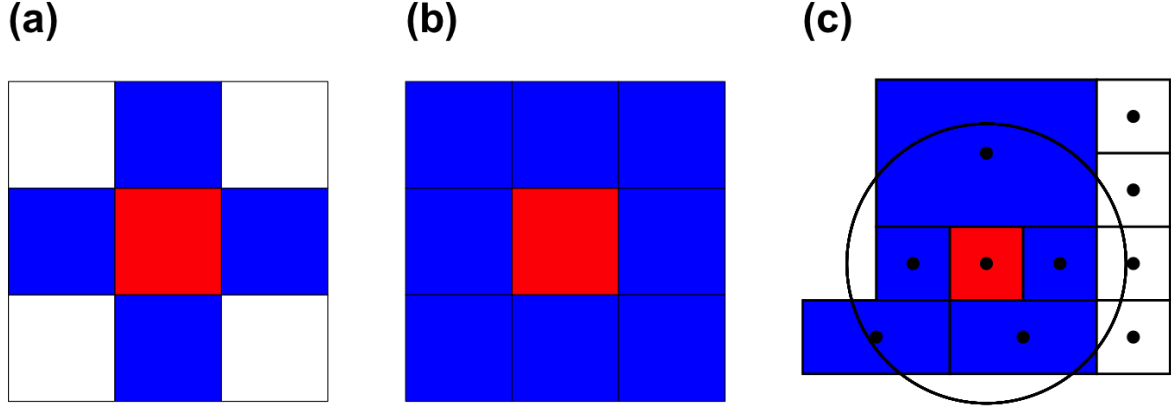


Figure 1: (a) Rook neighbors (blue) share an edge with a reference spatial region (red), (b) Queen neighbors (blue) share either an edge or vertex (or both) with a reference spatial region (red), and (c) Distance-based neighbors (blue) have centroids within a fixed radius of the reference cluster (red). Clusters with centroids beyond this distance are not considered as neighbors (white)[32]

2.2.2 Distance-Based

Distance-based neighbors[27] identify as neighbors any cluster whose centroid falls within a specified radius of a reference cluster (illustrated in Figure 1 (c)). A complete description of distance-based neighbor construction and the corresponding spatial weights formulation is provided in Appendix F. The simulation study presented in this paper relies exclusively on Rook contiguity for neighbor definition.

2.3 Intervention & Spillover Effects

Two primary effect types are considered to understand how treatments affect subjects in clusters where they are applied and subjects in the surrounding areas. The intervention effect and spillover effect are two parameters that are estimated in this study. Spatial cluster randomized trials have subjects or regions that are assigned to receive an intervention and others assigned as controls that do not receive an intervention. Observation of the resulting outcomes allows for estimation of the direct intervention effect and indirect spillover effect on the study region.

2.3.1 Intervention Effect

The intervention effect (β) is the *direct* structural coefficient in the spatial autoregressive model of Section D.1: the change in a cluster's outcome attributable to its own receipt of the intervention, holding

the spillover indicator z_i and the spatial lag fixed. Estimation of this coefficient is possible due to the presence of clusters assigned as controls for comparison. The relevant intervention could be a vaccine to prevent infectious disease, a training program for government employees, an educational program for students, or many others. We stress that β is a structural model coefficient and not a marginal potential-outcome contrast; this distinction is developed formally in the “Estimands and Interpretation” paragraph of Section D.1. When spillover is permitted to propagate back into intervention clusters (the TrtSpill mechanism; see Section 2.3.3), β continues to denote the direct coefficient while the spillover received by treated units is carried by the separate coefficient ψ on z_i . If an analyst were instead to fit a model that *omits* z_i , the coefficient on the treatment indicator would conflate the direct and indirect effects through omitted-variable bias—a phenomenon we characterize quantitatively in Section E and Section 4.2 (R2.M2).

2.3.2 Spillover Effect Methods

The spillover effect (ψ) represents an indirect effect on clusters where an intervention was not directly applied[33][34]. There are many to model a spillover effect as it is observed due to spatial proximity to subjects or regions where an intervention is applied. In this study, only binary spillover effects were considered. A binary spillover effect is a fixed level of a spillover effect that is applied if a subject or region is deemed to be a neighbor of a subject or region where the intervention is applied.

2.3.3 Spillover Effect Types

In this study, there are two specific ways in which spillover of the applied intervention is allowed to propagate to other cells or subjects. In both cases, spillover effects originate from areas where the original intervention treatment is applied. The first case allows for spillover to propagate from cells receiving the intervention into only cells assigned as control cells. In this case, the effect of the intervention is not permitted to spillover into other intervention cells. This suggests a case where the intervention effect applied is the maximum observable effect and any potential spillover cannot increase the observed effect of a particular intervention. Another method of spillover application is considered where spillover effects are permitted to be applied to both control AND intervention cells that are neighbors of a cell that receives an intervention. This application suggests that an effect greater than that of the original intervention applied can potentially be observed in a cell that received an intervention and also had

spillover from another intervention cell. It is also to be expected that any change in the estimated effects would not be observed in spatial settings where the treatment allocation does not satisfy block stratification conditions (where no intervention cells are rook neighbors with other intervention cells). Figure 2 provides a comparative representation of how these two methods permit the application of spillover effects on clusters based on their treatment assignment. In all scenarios of this simulation study, the application of any spillover effect is simply a binary indication of whether a cluster receives a spillover effect or not. Even if spillover exposure could potentially be directed from multiple neighboring clusters, the modeling in this study does not consider any additive spillover effects.

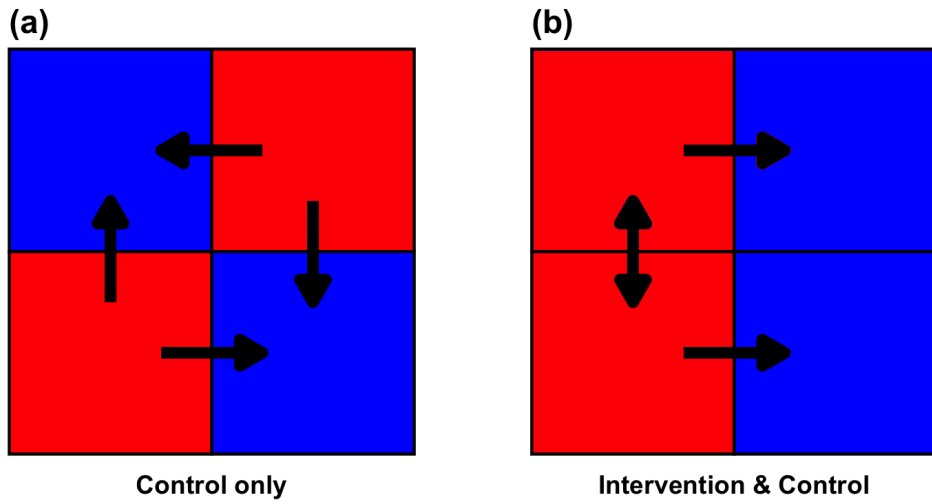


Figure 2: (a) TrtNoSpill: Spillover arrows flow *outward* from intervention cells (red) into neighboring control cells (blue) only; no spillover enters other intervention cells. (b) TrtSpill: Spillover arrows flow from intervention cells (red) into *both* neighboring control cells (blue) and neighboring intervention cells (red).

2.4 Simple Random vs. Block Stratified Sampling

In this study, two different treatments are considered: intervention & control. Assignment of these treatments is considered under two sampling methods: Simple Random Sampling (SRS) & Block Stratified Sampling (BSS). Random sampling treatment assignment considers all possible combinations of intervention & control arrangements on a specified spatial grid and provides equal probability of assignment for any given combination of intervention and control. For example, a 2 row by 3 column spatial grid has 6 total cells. Consideration of a 1:1 ratio between intervention and control treatments provides that three

cells are assigned as intervention and three cells are assigned as control. Therefore, there are ${}_6C_3 = 20$ different combinations of treatments for this specific spatial region.

Block stratified sampling ensures a balanced number of intervention and control cells are selected, with the added condition that no intervention or control cells share a common edge boundary. This approach is analogous to geographically stratified randomization in standard CRTs, where spatial proximity is used to define strata rather than demographic or clinical covariates[35][36][3]. The added spatial isolation condition—that no two same-treatment clusters may share a Rook edge—is a novel constraint introduced in this work to directly minimize cross-cluster spillover at the design stage. An exception to this rule occurs in spatial setups where both dimensions are odd. In this case, the count of intervention and control cells must be out of balance and we assert that the control class will always be in the majority. In these instances, we relax the common edge boundary restriction for control cells and only require that intervention cells not share a common edge boundary in order to satisfy the block stratification condition.

We now consider a series of spatial grid setups to address efficient estimation of the intervention effect. These spatial grids are have dimensions denoted as (row $\times$ column) and we seek to apply a balanced distribution between intervention and control cells where possible. When this isn't possible, we defer to the control group being the majority class as mentioned previously.

2.4.1 Grid geometries and allocation counts

We consider three spatial grid geometries, denoted (row $\times$ column): a 2×4 grid (8 cells), a symmetric 3×3 grid (9 cells), and a 3×4 grid (12 cells). For each, simple random sampling (SRS) draws from all balanced (or, for the odd-dimensioned 3×3 grid, near-balanced) intervention/control arrangements, whereas block stratified sampling (BSS) retains only those arrangements satisfying the spatial isolation condition. Table 1 summarizes the resulting allocation spaces, and Figure 6 shows a representative BSS (top row) and SRS (bottom row) allocation for each grid. The spatial objects are constructed with the Simple Features for R package[37][30].

3 Application

The findings from our simulation study provide relevant insights for investigators designing CRTs in real-world applications where spatial dependence and spillover effects are present and must be accounted for. In this case, a pair of studies in development that we consider involve the North Carolina Department of Adult Correction. The structure of correctional systems, particularly community supervision, often involves geographically clustered units where staff from different clusters may interact and regional similarities can influence outcomes. The investigators leading these studies desire recommendations for treatment assignment methodology regarding the spatially clustered nature of their design.

3.1 Background of Probation in North Carolina

The North Carolina Department of Adult Correction Division of Community Supervision (NC DCS) provides a clear example of a spatially structured organization, making it an ideal context for applying our research. The NC DCS is organized into four geographic Divisions numbered 1 through 4 from East to West, which are subdivided into 30 Districts that make up the entirety of the state. Each district is composed of one or more counties. If there is more than one county in the district, the counties are adjacent, but some districts are composed of only a singular county. In these districts, Probation and Parole Officers (PPOs) supervise individuals at the county level and are themselves managed by Chief Probation/Parole Officers (CPPOs) and Judicial District Managers (JDMs) who oversee districts. Each division (composed of multiple districts) is managed by a Judicial Division Director (JD Dir) as well. A map of the segmentation of each division and district is presented in Figure 3.

This geographic and administrative structure means that Chiefs and Judicial District Managers (JDMs) often span multiple units and counties, creating a direct channel for spillover effects among regions under their supervision. For instance, if leadership views an intervention in one county favorably, they may implement its components in other counties under their jurisdiction, even if those areas were not assigned the intervention. Furthermore, adjacent counties within the same region often share demographic characteristics, labor market conditions, and similar judicial cultures, as well as the same managed care organizations and treatment providers, leading to spatial dependence in probation outcomes. The choice of treatment assignment sampling strategy in this context is non-trivial when considering potential options of either simple random sampling (SRS) or block-stratified sampling (BSS).

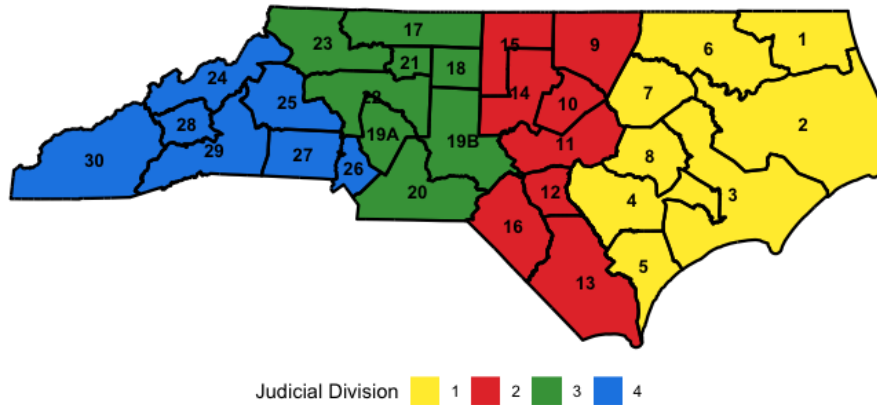


Figure 3: The North Carolina Department of Adult Correction Division of Community Supervision map is segmented into 30 judicial districts that are grouped into 4 judicial divisions.

Selection of the appropriate treatment sampling strategy may yield beneficial downstream outcomes when considering the error of intervention & spillover estimates.

3.2 Case 1: Trauma-Informed and Engaged Supervision

One proposed study involves implementing a trauma-informed supervision model for PPOs managing specialty caseloads like domestic violence, sex offender, or gang affiliated cases. The intervention consists of intensive training and follow-up sessions delivered to cohorts of PPOs and CPPOs. The primary outcomes would be changes in officer knowledge and attitudes, as well as impacts on individuals under their supervision. The key design decision for this study is the unit of randomization. Randomizing at the PPO level is not feasible due to the high likelihood of spillover within a county office. Therefore, a cluster randomized design at the county or district level is necessary. This cluster randomization decision is also supported by agency-related logistics in addition to considerations for spillover. Original plans for a similar study, acknowledging the risk of contamination between adjacent units, considered randomization at the district level specifically to minimize spillover. This scenario directly reflects the conditions modeled in our simulation, where the effectiveness of an intervention in one cluster may influence outcomes in neighboring control clusters where an intervention is not applied.

3.3 Case 2: Testing an implementation strategy for specialty mental health probation

Another study proposal aims to test a strategy for improving the implementation of Specialty Mental Health Probation (SMHP). The intervention involves enhancing collaboration between PPOs and community-based behavioral health service providers through cross-training sessions and joint treatment team meetings. The primary outcome is improved collaboration, with secondary outcomes related to treatment engagement for individuals on probation. This case presents a more complex spatial challenge. In addition to the geographic proximity of probation districts, the behavioral health system in North Carolina is managed by four large Managed Care Organizations (MCOs), each responsible for a different coverage area. Some service provider agencies are large, operating in multiple counties and potentially across different MCO regions. This creates multiple, overlapping layers of potential spillover. Figure 4 presents a map of which counties in the state that each MCO operates. An intervention implemented in one county could be adopted by a provider and informally carried into a neighboring control county, or an MCO could promote a similar initiative, confounding the study's results. This complex spatial linkage underscores the importance of a robust treatment assignment strategy in order to efficiently estimate the impact of the intervention.

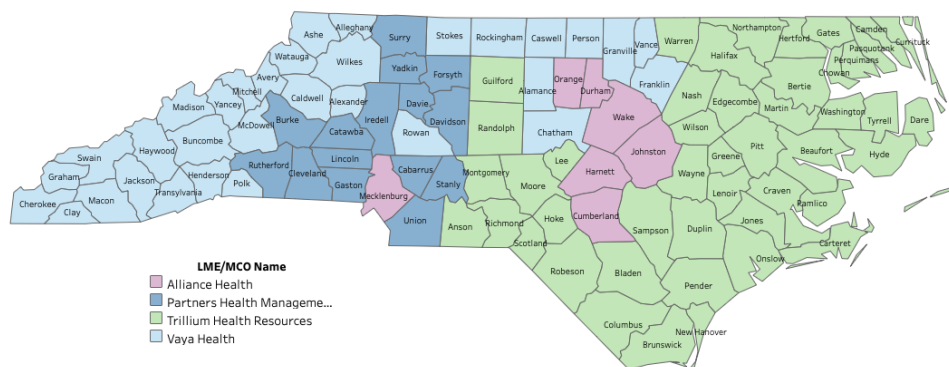


Figure 4: Map of Local Management Entity/Managed Care Organizations (LME/MCOs) for each county in the state of North Carolina. There are 4 LME/MCOs of which one operates in each of the 100 counties of the state.

3.4 Recommendations for treatment assignment

The results of our simulation study indicate a clear trade-off: Simple Random Sampling (SRS) reduces the average mean squared error (MSE) across many randomizations (intervention/control combinations), while Block Stratified Sampling (BSS) reduces the variance of the error although it is not minimized. Put simply, BSS is less likely to produce a “perfect” assignment by chance but is also robust against a “worst-case” assignment that could render the study results invalid. We recommend BSS for both case studies presented, but with an important and explicit qualification: this recommendation rests on the expectation that spillover in these correctional-supervision settings flows primarily from intervention units into neighboring control units (the TrtNoSpill regime), where BSS reliably controls estimation variance. If investigators instead anticipate substantial bidirectional spillover among treated units together with background spatial dependence, our simulations show that the spatial separation enforced by BSS can amplify rather than reduce bias (the checkerboard paradox, Section 4); in that regime SRS, or an analysis-stage buffer-zone design, may be preferable. The recommendation below is therefore conditional on the anticipated spillover mechanism, and we encourage investigators to state that assumption explicitly when justifying their design.

In Case 1 (Trauma-Informed Supervision), a simple random assignment of districts could, by chance, result in a geographic imbalance, concentrating treated districts in one part of the state. This would confound the intervention effect with regional characteristics. A more robust design would be to stratify by the four state divisions and randomize districts to intervention and control conditions within each division with a block stratification strategy. This ensures geographic balance and leverages the finding that BSS minimizes estimation variance, providing a more stable and defensible estimate of the intervention effect.

In Case 2 (SMHP Implementation), the rationale for BSS is even stronger due to the complex potential for spillover. A simple random assignment of counties would ignore the significant structural influence of the MCOs. The most effective design would be to stratify by MCO coverage area, randomizing counties to the implementation strategy (intervention) or control condition within each MCO. This approach directly controls for the significant confounding variable of MCO-level policies and provider networks. However, some challenges with this MCO stratification do persist. For example, it’s possible that a county is selected for the intervention and is supervised by a CPPO or even a judicial district manager

that oversees a control group county. Furthermore, multiple service providers may serve a given county in the coverage area of an MCO and it is possible that multiple service providers begin team meetings. Another consideration is that some service providers are very large and have offices around the state and contract with more than one MCO. Based on this, we must consider potential spillover if after participating in a treatment team meeting, a service provider intends initiate those practices in another part of the state. Given the high risk of contamination, the robustness of BSS in minimizing error variance makes it a prudent choice for achieving a reliable effect estimate, provided spillover is expected to flow chiefly into control units. While SRS might achieve a lower MSE (intervention error estimate) in theory, the risk of a single, poorly distributed randomization in a complex, real-world setting outweighs the potential benefit. We note one caveat specific to this case: because the SMHP setting carries a genuine possibility of bidirectional spillover among treated units (e.g., a large provider carrying practices across MCO regions), investigators should weigh whether an analysis-stage buffer-zone exclusion of the most contamination-prone boundary counties would further protect the estimate—an option that, as discussed in Section 4, directly addresses the identifiability limitation rather than only its variance. Subject to that consideration, it is wise to proceed with a sampling strategy that manages estimation error with minimal variation.

A representation of this block stratified sampling strategy where we “block” treatment and control districts together with an effort to isolate districts of a particular intervention status from each other is shown in Figure 5. Note that across the entire map of the state and specifically in Judicial Division 1 (easternmost region) that when possible districts of the same intervention status are isolated from each other. When districts share a border with multiple other districts, this sampling method ensures that treatment districts are isolated from other treatment districts as best as possible. It is important to note that, unlike the idealized rectangular grids examined in the simulation study, real-world jurisdictions have irregular shapes and varying numbers of shared borders. The allocation depicted in Figure 5 therefore represents a *best-effort* block stratification: the assignment minimizes shared borders between same-treatment districts while acknowledging that perfect spatial separation cannot always be achieved in practice. Investigators applying BSS to real-world geographic settings should expect some deviation from the strict no-common-edge criterion and should document the degree to which the chosen allocation approximates the ideal.

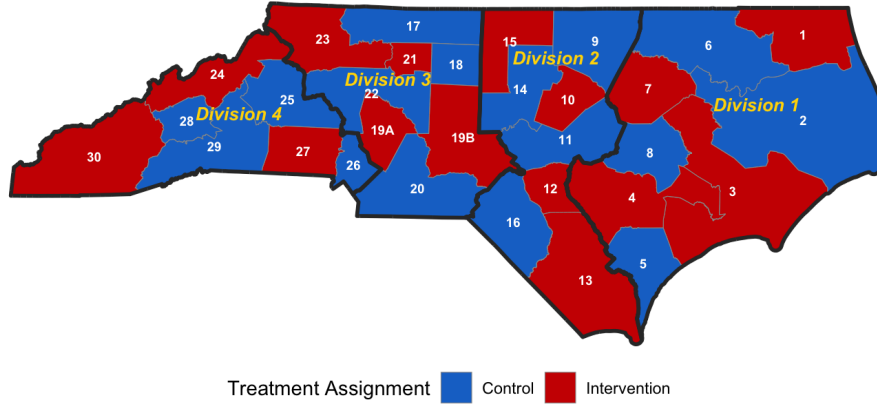


Figure 5: Map of North Carolina with example block stratified sampling intervention assignments for each judicial district under consideration. Districts are assigned to either treatment (red) or control (blue).

4 Discussion

4.1 Relevance of results

This simulation study provides insights into the efficiency of treatment assignment methodologies in spatial cluster randomized trials (CRTs) when considering spillover effects and spatial dependence. A primary contribution of this work is the direct comparison between Simple Random Sampling (SRS) and Block Stratified Sampling (BSS) across various scenarios, offering practical guidance for investigators designing such trials.

The spatial separation principle underlying BSS has antecedents in spatially balanced experimental designs. For example, McCann et al. (2018) demonstrated that spatially stratified allocation in field trials reduces confounding by ensuring treatment balance across the spatial domain[35]. However, the formal comparison of SRS and BSS within the context of spatial cluster randomized trials that incorporate both a spatial autoregressive (SAR) model for background dependence *and* an explicit spillover mechanism represents a novel contribution of the present work. To our knowledge, no prior study has examined how the choice between these two allocation strategies affects the bias–variance decomposition of the intervention effect estimator under SAR modeling with simultaneous spillover effects, nor has the

exhaustive enumeration of all feasible allocations been used to characterize the full design space of a spatial CRT.

A related line of work concerns covariate-constrained randomization, in which the set of eligible allocations is restricted to those achieving covariate balance across treatment arms[36]. While BSS and covariate-constrained randomization both restrict the allocation space, they operate on fundamentally different principles: BSS imposes a spatial isolation condition based on cluster adjacency (no two same-treatment clusters share a Rook boundary), whereas covariate-constrained randomization restricts allocations based on balance of measured covariates. These approaches are complementary and could in principle be combined; however, the present work focuses on the spatial isolation constraint as the design criterion, leaving covariate-balance extensions to future research.

As detailed in Section E, three patterns emerge consistently across grids. First, increasing the spillover magnitude ψ raises the MSE of $\hat{\beta}$ regardless of design, underscoring the importance of accounting for spillover at the design stage. Second, even slight background spatial dependence ($\rho = 0.01$) inflates MSE for both methods. Third, and most consequential for design choice, the two methods embody a bias–variance trade-off whose direction depends on the spillover regime: under control-only spillover (Trt-NoSpill) BSS yields low-variance and typically low-MSE estimates, whereas under bidirectional spillover (TrtSpill) SRS is usually more accurate on average. SRS thus provides a lower-bias central estimate while BSS provides greater consistency across the possible allocations—a contrast whose practical implications we develop below.

It is also insightful to note that for each simulation grid setup (2×4 , 3×3 , and 3×4), there are 7 different treatment assignment combinations that are the overall “optimal” combination for at least one of the simulation scenarios in terms of MSE of the intervention effect estimate. It appears that the optimal treatment combination shifts as the components of each simulation scenario are adjusted and there is no consistent treatment combination from the Simple Random Sampling strategy that is optimal. This consistency is the principal appeal of BSS: it furnishes a low-variance estimate that is never the single best case but, importantly, is also insulated from the worst-case allocations that SRS can draw by chance. We are careful, however, not to over-generalize this into a blanket recommendation. As the results make clear, BSS is advantageous chiefly when spillover flows only into control clusters (Trt-NoSpill) and is *not* uniformly preferable: under bidirectional spillover combined with spatial dependence (TrtSpill,

$\rho > 0$) the checkerboard structure can make BSS the worst-performing design (the checkerboard paradox, below). The recommendation between SRS and BSS is therefore conditional on the anticipated spillover regime rather than universal.

The magnitude of the bias warrants candid discussion. With a true intervention effect of $\beta = 1.0$ and spillover magnitudes $\psi \in [0.5, 0.8]$, the bias of $\hat{\beta}$ under exactly collinear allocations equals $-\psi$ (Section E), i.e. 50–80% of the true effect—a distortion large enough to reverse practical conclusions. We therefore agree that the estimation error can be severe. Crucially, however, this severity is not an indictment of every design: it is a direct, predictable consequence of the exact collinearity between the treatment and spillover indicators that arises when an allocation surrounds every control with treated neighbors (and conversely). The bias is largest precisely where identifiability is lost (BSS under TrtNoSpill, and the checkerboard under TrtSpill+ ρ) and shrinks toward zero where the allocation permits the spillover term to be separately estimated (many SRS allocations under TrtSpill, where SRS bias is on the order of a few percent of β ; see Table 3). The actionable message is thus not “no design works” but rather that naïve estimation under spatial confounding is dangerous and that design choice interacts strongly with the identifiability of the spillover effect.

An important nuance observed in the 3×4 simulation results warrants attention in the context of practical design recommendations. When BSS forces an exact checkerboard arrangement—maximally separating intervention and control clusters—every intervention cluster is surrounded exclusively by control neighbors. Under TrtSpill, this means spillover flows from each intervention cluster into its control neighbors *and* back into neighboring intervention clusters (which are diagonally adjacent but not Rook neighbors). This structural property creates a systematic pattern of amplified spillover-induced bias specifically for BSS in the TrtSpill regime with spatial dependence, producing estimation error that exceeds that of SRS in that scenario. We refer to this as the “checkerboard paradox”: the very design feature that makes BSS effective at minimizing spillover contamination under TrtNoSpill—complete spatial separation of treatment arms—becomes a liability when bidirectional spillover is expected. Investigators should therefore consider their prior knowledge of the spillover mechanism when selecting between SRS and BSS, with BSS most advantageous when spillover is expected to flow primarily into control clusters.

In summary, this study offers a comparison of two opposing sampling methodologies in spatial CRTs.

The patterns observed regarding spillover magnitude, spatial dependence, and spillover application types provide practical information for investigators. They must consider the goal of an unbiased average estimate (favored by SRS) versus the need for consistent, low-variance estimates (achieved with BSS), especially when designing studies where spillover effects are expected.

4.2 Limitations

While this simulation study provides valuable insights, it is important to acknowledge a few key limitations that may affect the generalizability and scope of its findings. The study encountered issues related to multicollinearity between the direct (intervention) and indirect (spillover) effects. In some spatial configurations and with specific spillover magnitudes, high correlation between the x_i (intervention indicator) and z_i (spillover indicator) variables in the spatial autoregressive model (Equation 9) could lead to instability in parameter estimates. While the SAR model addresses spatial relationships, extreme collinearity can still impact the precision of individual coefficient estimates, potentially affecting MSE calculations. We emphasize, however, that the most consequential manifestation of this issue in our results is not merely imprecision but systematic *bias*: as derived in Section E, when the allocation makes z_i an exact function of x_i (every BSS/TrtNoSpill configuration), β and ψ are not separately identifiable and $\hat{\beta}$ estimates $\beta - \psi$, yielding a bias of $-\psi$. This is a structural identification limitation of compact, fully separated allocations on small grids, not a numerical artifact, and it is the mechanism behind the large BSS biases reported here. This issue in separating highly correlated effects is known in spatial causal inference and requires careful consideration in applied settings; analysis-stage buffer-zone designs (Section 4, Future Considerations) are one principled remedy because they restore identifiability by construction. As discussed in Section D.1, ρ (background spatial autocorrelation) and ψ (causal spillover) are conceptually distinct but both depend on the same neighborhood structure, creating an identification challenge when both are large; in such settings, estimates of $\hat{\psi}$ and $\hat{\rho}$ may exhibit instability, inflating MSE for both parameters. It is also necessary to note that the expanded simulation space quickly becomes computationally infeasible. The current study explored a specific range of spillover effect sizes, spatial dependence parameters, and grid dimensions. However, a more extensive exploration of additional parameters (e.g., different baseline effects, varying intervention effect sizes, or a wider range of spatial dependence values) or more complex spatial structures would require significantly greater computational

resources. This limitation restricted the number of scenarios that could be thoroughly investigated within the current framework.

An additional notable limitation is the rudimentary spillover mechanism (binary effect) applied. This simulation study modeled spillover as a fixed, binary event, meaning a unit either experienced the full potential spillover effect or none, based solely on contiguity (Rook neighbors). This approach does not account for more realistic scenarios where spillover effects may decay with distance, or where cumulative exposure from multiple treated neighbors could lead to additive or non-linear effects. The absence of more sophisticated distance-based decay functions (e.g., linear, categorical, exponential, or Gaussian decay) or mechanisms for additive spillover limits the direct applicability of these specific quantitative results to real-world situations where such nuanced spillover dynamics are likely. A further limitation is the cluster-level uniformity of spillover exposure. The current simulation assigns binary spillover at the cluster level: all subjects within a cluster either experience or do not experience spillover based solely on whether their cluster shares a Rook boundary with an intervention cluster. In practice, subjects near the boundary of a control cluster may experience stronger spillover than those near the center, while subjects near non-intervention boundaries may experience none at all. This within-cluster heterogeneity in spillover exposure is not captured by the current model, and future work could explore boundary-level or subject-level spillover mechanisms that account for the spatial position of subjects within clusters.

Because this study evaluates estimator properties through exhaustive enumeration of all possible treatment allocations, the findings characterize the *design-space* performance of each sampling strategy rather than the sampling distribution arising from a single randomization. The simulation is not designed to support permutation-based inference under BSS; model-based inference via SAR MLE is the appropriate framework for practitioners, and the simulation results should be interpreted as describing how well each design tends to perform across the full space of feasible allocations for a given grid geometry. Furthermore, adopting BSS as the randomization mechanism in a real-world trial would yield a near-deterministic allocation with only 2–6 feasible arrangements. Permutation-based randomization tests under BSS are therefore infeasible in practice; investigators implementing BSS should rely on model-based (SAR MLE) inference and should present this choice explicitly when reporting their study design.

The current model also did not consider higher-order neighbors beyond immediate (Rook) contiguity.

The study also utilized homogeneous cluster sizes and regular grid shapes. Real-world clusters often exhibit irregular shapes and varying sizes, which could influence spillover dynamics and the efficiency of different sampling strategies. The current findings may not fully extend to such heterogeneous spatial arrangements. Furthermore, the study assumes a fixed treatment allocation ratio (primarily 1:1 or near 1:1). Varying the proportion of intervention to control clusters could impact the statistical power and bias of intervention effect estimates, especially in the presence of spillovers. Finally, it is crucial to note that the results obtained and recommendation for a Block Stratified Sampling (BSS) strategy hold under the conditions of the spatial autoregressive (SAR) model and it cannot be guaranteed that the apparent benefits of BSS persist under the conditions of an alternative spatial model.

4.3 Future Considerations

Building on these findings, several extensions would sharpen design guidance for spatial CRTs. First, richer spillover structures beyond the binary mechanism—distance-based decay (linear, step-wise, exponential, or Gaussian) and *additive* exposure that accumulates with the number of treated neighbors—would better reflect how indirect effects attenuate over space and how cumulative contamination behaves. Second, alternative neighbor definitions (Queen or higher-order contiguity) together with irregular, heterogeneously sized clusters and non-uniform within-cluster subject density would test how robust the SRS/BSS contrast is to realistic geography. Third, broadening the parameter sweep (baseline and intervention effect sizes, a wider range of ρ , and unbalanced allocation ratios) and assessing robustness to model misspecification—for example fitting a Spatial Error or Spatial Durbin model, or generating data from a process that departs from the SAR form—would map the conditions under which each design is preferable and clarify the implications for statistical power and cost-effectiveness.

A natural and important extension, suggested by the Reviewer, is to evaluate buffer-zone (“fried-egg”) designs, in which boundary clusters between treatment arms are excluded from the primary analysis to reduce spillover contamination[8]. Such designs are an especially promising response to the identifiability problem highlighted in this work: by excluding the control clusters that border treated clusters, a buffer-zone design restores $z_i = 0$ for the analyzed controls, which breaks the exact $x_i - z_i$ collinearity *by design* and would permit β and ψ to be separately identified rather than conflated. In this sense buffer zones address the root cause of the large biases reported here, whereas BSS and SRS manage

them only through the variance–bias trade-off. Buffer zones thus represent a complementary, analysis-stage strategy to the allocation-stage approaches studied here: rather than selecting an allocation that minimizes cross-boundary spillover, they remove the clusters most likely to be contaminated. Characterizing the comparative performance of BSS, SRS, and buffer-zone exclusion strategies—particularly in the TrtSpill + spatial-dependence scenarios where BSS showed amplified bias—would provide more complete guidance for practitioners and is an important next step in this line of research.

Declarations

Ethics Approval and Consent to Participate

Not applicable. This study involves simulation data only and does not require ethics approval or consent.

Consent for Publication

Not applicable.

Availability of Data and Materials

All data supporting the findings of this study are generated via simulation for the included simulation study. The simulation data and R code are available upon request and with permission of the authors.

Competing Interests

The authors declare that they have no competing interests.

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Authors' Contributions

A.W. conceptualized and designed the simulation study, wrote the manuscript text, and prepared Figures 1–7 and 10. T.V. provided Figures 8 and 9 and contributed advisement for Section 4 of the manuscript. F.-C.L. provided methodological guidance and critical review throughout. All authors reviewed and approved the final manuscript.

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Appendices

The following appendices contain technical material moved from the main text to improve readability.

Each appendix is cross-referenced from the relevant section of the main manuscript.

A Spatial Weights Matrix Formulation

The spatial weights matrix $\mathbf{W}$ is a cluster-level $n \times n$ matrix encoding the presence and strength of spatial connections between clusters i and j . Two types of weights are considered in this work.

Indicator-Based (Contiguity) Weights

Raw indicator-based weights are defined as

$$\tilde{w}_{ij} = \begin{cases} 1, & \text{if clusters } i \text{ and } j \text{ are Rook neighbors (shared edge)} \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

and are row-normalized to give the entries

$$w_{ij} = \frac{\tilde{w}_{ij}}{\sum_{j \neq i} \tilde{w}_{ij}} \quad (2)$$

so that each row of $\mathbf{W}$ sums to 1, making the spatial lag $\rho \mathbf{W} \mathbf{y}$ a weighted average of neighboring clusters' responses. For Rook contiguity this yields $w_{ij} = 1/|\mathcal{N}(i)|$ for each of cluster i 's $|\mathcal{N}(i)|$ Rook neighbors and 0 otherwise, matching the inline definition given after Equation (9). It is these row-normalized entries w_{ij} that appear throughout the main-text model equations.

Distance-Based Weights

Distance-based weights assign strength proportional to the inverse distance between cluster centroids:

$$\tilde{w}_{ij} = \begin{cases} \frac{1}{d_{ij}}, & \text{if } 0 < d_{ij} < d_{\max} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where d_{ij} is the distance between the centroids of clusters i and j and $d_{\max}$ is the maximum distance threshold for neighborhood inclusion. These weights are also row-normalized via

$$w_{ij} = \frac{\tilde{w}_{ij}}{\sum_{j \neq i} \tilde{w}_{ij}}. \quad (4)$$

In this study, indicator-based (Rook contiguity) weights are used throughout all simulation experiments.

B SAR Model: Full Likelihood

The subject-level spatial lag model is

$$y_{ik} = \rho \sum_{j \neq i} w_{ij} y_{jk'} + \alpha + \beta x_i + \psi z_i + \epsilon_{ik} \quad (5)$$

with residuals

$$\epsilon_{ik} = y_{ik} - \rho \sum_{j \neq i} w_{ij} y_{jk'} - (\alpha + \beta x_i + \psi z_i). \quad (6)$$

Assuming $\epsilon_{ik} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, the likelihood function is

$$L(\alpha, \beta, \psi, \rho, \sigma^2) = \frac{|\mathbf{I} - \rho \mathbf{W}|}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \boldsymbol{\epsilon}^\top \boldsymbol{\epsilon}\right) \quad (7)$$

where $|\mathbf{I} - \rho \mathbf{W}|$ is the Jacobian of the transformation from $\boldsymbol{\epsilon}$ to $\mathbf{y}$. The log-likelihood is

$$\begin{aligned} \log L(\alpha, \beta, \psi, \rho, \sigma^2) &= \log |\mathbf{I} - \rho \mathbf{W}| - \frac{nm}{2} \log(2\pi\sigma^2) \\ &\quad - \frac{1}{2\sigma^2} \sum_{i=1}^n \sum_{k=1}^m \left(y_{ik} - \rho \sum_{j \neq i} w_{ij} y_{jk'} - \alpha - \beta x_i - \psi z_i \right)^2 \end{aligned} \quad (8)$$

which is maximized with respect to $\alpha, \beta, \psi, \rho$, and σ^2 , over n clusters each containing $m = 20$ subjects.

In practice, this maximization is performed by the `lagsarlm()` function from the `spdep` R library[28].

C Simulation Algorithm

The step-by-step simulation procedure used to generate results for all spatial grid geometries and scenarios is as follows.

1. **Define scenario.** Specify the spatial grid geometry and fix the simulation parameters ($\alpha = 0.2$, $\beta = 1.0$, $\psi \in \{0.5, 0.6, 0.7, 0.8\}$, $\rho \in \{0.00, 0.01\}$) along with the spillover application type (TrtNoSpill or TrtSpill).

2. **Enumerate treatment combinations.** List all possible combinations of intervention and control cluster assignments for the specified grid: 70 for the 2×4 grid, 126 for the 3×3 grid, and 924 for the 3×4 grid.
3. **Construct spatial object.** Build the grid object (8, 9, or 12 clusters) using the `sf` R package[37] and define Rook neighbors for each cluster.
4. **Identify BSS allocations.** From the full set of combinations, identify those satisfying the block stratification condition: 2 for the 2×4 grid, 6 for the 3×3 grid, and 2 for the 3×4 grid.
5. **Simulate for each combination.** For each treatment combination:
 - (a) Sample 20 subjects per cluster with uniformly distributed coordinates over the spatial extent (160, 180, or 240 subjects total).
 - (b) Assign each subject to its cluster and intervention/control status.
 - (c) Determine spillover eligibility: for each subject, set $z_i = 1$ if the subject's cluster has at least one Rook neighbor assigned to the intervention and the spillover application type permits it; otherwise $z_i = 0$.
 - (d) Generate outcomes from the closed-form SAR data-generating process: $\mathbf{y} = (\mathbf{I} - \rho \mathbf{W})^{-1}(\boldsymbol{\alpha} + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon})$, where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$.
 - (e) Fit the spatial lag model via `lagsarlm()` and retain estimates $(\hat{\alpha}, \hat{\beta}, \hat{\psi}, \hat{\rho})$.
 - (f) Repeat (a)–(e) $N = 50$ times for the 2×4 and 3×3 grids; $N = 10$ times for the 3×4 grid.
6. **Aggregate estimates.** For each treatment combination, compute the mean, variance, bias, and MSE of $\hat{\beta}$ (and other parameters) across the N replications.
7. **Summarize by design.** Aggregate metrics across all SRS combinations and separately across all BSS combinations to produce the summary statistics reported in Tables 1–3.
8. **Repeat.** Repeat Steps 1–5 for each grid geometry (2×4 , 3×3 , 3×4).

D Detailed Simulation Results

The following tables report the full intervention-effect ($\hat{\beta}$) error estimates—MSE, standard deviation, and bias—for SRS and BSS across all three grid geometries, separated by spillover regime. Table 2

727 covers control-only spillover (TrtNoSpill) and Table 3 covers bidirectional spillover (TrtSpill). These
 728 tables provide the numeric detail underlying the figures and summary statements of Section E.

Grid	Cells	Intervention : Control	SRS allocations	BSS-valid allocations
2×4	8	4 : 4	${}_8C_4 = 70$	2
3×3	9	4 : 5	${}_9C_4 = 126$	6
3×4	12	6 : 6	${}_{12}C_6 = 924$	2

Table 1: Treatment-allocation counts for each spatial grid. SRS enumerates all balanced arrangements; BSS retains only spatially isolated (“checkerboard”) allocations in which no two same-treatment clusters share a Rook edge. The 3×3 grid has odd dimensions, so an exact balance is impossible: the control class is held in the majority (5:4) and the no-shared-edge condition is relaxed for control clusters, yielding one strict plus five relaxed BSS allocations (6 total).

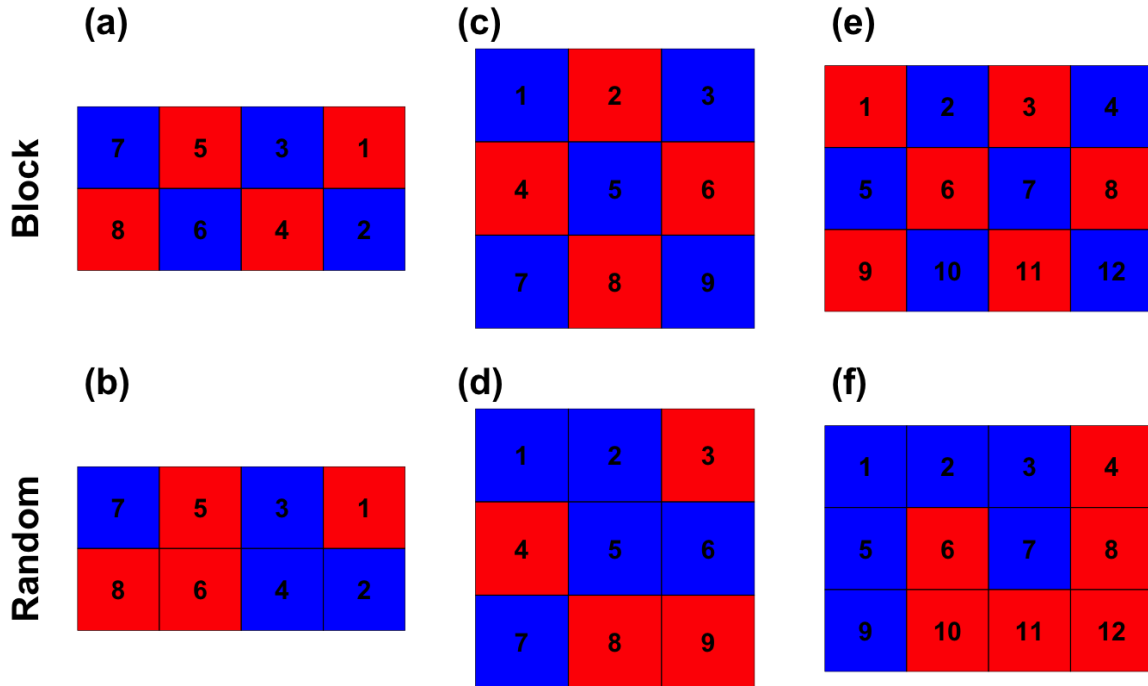


Figure 6: 2x4 (left), 3x3 (middle), and 3x4 (right) spatial grids of intervention (red) and control (blue) clusters assigned via Block Stratified Sampling (top row) and Random Sampling (bottom row)

729 The salient feature is that BSS collapses an allocation space of 70–924 arrangements down to just 2–6.
 730 For the even-celled 2×4 and 3×4 grids this forces an exact checkerboard; on the 3×4 grid in particular

every intervention cluster is then surrounded entirely by control neighbors, a structural property that proves consequential under bidirectional spillover (Section E). This drastic reduction is the source of both BSS’s principal strength—near-zero variance across its handful of allocations—and its principal limitation: a near-deterministic design that cannot support randomization-based inference (Section E, Section 4.2).

D.1 Spatial Autoregressive (SAR) Model

Outcomes in the presence of spatial dependence and spillover effects can be considered via modeling with a spatial autoregressive model [38][27][39]. A spatial autoregressive model has the form:

$$y_{ik} = \rho \sum_{j \neq i} w_{ij} y_{jk'} + \mathbf{x}_i \beta + \varepsilon_{ik} \quad (9)$$

where w_{ij} are elements of the row-normalized spatial weights matrix $\mathbf{W}$ (equal to $1/|\mathcal{N}(i)|$ if clusters i and j share a Rook edge, and 0 otherwise; full construction in Appendix A); i indexes clusters ($i = 1, \dots, n$) and k indexes individual subjects within cluster i , so the subscripts in $w_{ij} y_{jk'}$ denote cluster j and an individual subject k' in that cluster; ρ is the spatial correlation (autoregressive) coefficient; and ε_{ik} represents an i.i.d. error term for subject k in cluster i . The matrix $\mathbf{W}$ is a cluster-level $n \times n$ adjacency matrix—the subscripts i and j in w_{ij} refer to clusters, not individuals. This can also be expressed in matrix notation as:

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \beta + \varepsilon \quad (10)$$

It follows that the general form of the response for this model for the purpose of this study is

$$y_{ik} = \alpha + \beta x_i + \psi z_i + \rho \sum_{j \neq i} w_{ij} y_{jk'} + \epsilon_{ik} \quad (11)$$

where y_{ik} is the response for individual k in cluster i ; α is the baseline effect; β is the *direct* intervention effect; $x_i \in \{0, 1\}$ is the cluster-level treatment indicator (1 if cluster i receives the intervention, shared by all subjects k within that cluster); ψ is the scalar spillover effect parameter; $z_i \in \{0, 1\}$ is the cluster-level spillover indicator (1 if cluster i is a Rook neighbor of at least one treated cluster and the spillover type permits it; 0 otherwise); and the spatial lag $\rho \sum_{j \neq i} w_{ij} y_{jk'}$ captures background spatial dependence

through the weighted average of neighboring cluster outcomes (equivalently $\rho \mathbf{W}\mathbf{y}$ in matrix form). Because x_i and z_i are cluster-level quantities constant for all subjects within cluster i , all individuals in a cluster share the same expected outcome given their cluster assignment; individual responses differ only through the error term ϵ_{ik} . The spatial lag is built from ρ , the spatial autocorrelation parameter; $\mathbf{W}$, the row-normalized cluster-level spatial weights matrix (Appendix A); and $\mathbf{y}$, the full response vector. Finally, ϵ_{ik} is the i.i.d. error term for subject k in cluster i .

It is important to clarify the conceptual distinction between two spatial quantities in this model: the spatial autocorrelation parameter ρ and the spillover effect ψ . The parameter ρ is a *nuisance* parameter reflecting background spatial correlation in outcomes—nearby clusters tend to produce similar outcomes due to shared environmental, demographic, or social factors, independent of any intervention. In contrast, the spillover effect ψ is a *causal, design-level* quantity representing the indirect effect that an intervention in one cluster exerts on outcomes in neighboring clusters. While both quantities involve spatial proximity, ρ is an intrinsic property of the data-generating process that must be accounted for in order to obtain valid inferences, whereas ψ is of direct scientific interest and often a primary estimand alongside the intervention effect β . Failure to distinguish these two quantities—or to include both in the model—can produce confounded estimates of the intervention effect. In particular, when spatial dependence is high, failing to include ρ in the model may cause the spillover estimate $\hat{\psi}$ to absorb background correlation, overstating the apparent indirect effect of the intervention.

Estimands and Interpretation. The quantities β , ψ , and ρ are *structural coefficients* of the spatial autoregressive model, not marginal potential-outcome contrasts. This distinction matters because the SAR specification in Equation (9) places the response on both sides of the equation, so the model’s reduced (equilibrium) form is

$$\mathbf{y} = (\mathbf{I} - \rho \mathbf{W})^{-1} (\alpha \mathbf{1} + \mathbf{X}\beta + \mathbf{Z}\psi + \boldsymbol{\epsilon}). \quad (12)$$

A change in any cluster’s treatment status therefore propagates to all clusters through the spatial multiplier $(\mathbf{I} - \rho \mathbf{W})^{-1}$. As a consequence, a potential-outcome indirect effect such as $E[Y(0, \mathbf{1})] - E[Y(0, \mathbf{0})]$ —the contrast between every other cluster being treated versus untreated, holding a focal cluster at control—is a *derived* quantity that depends jointly on β , ψ , ρ , and $\mathbf{W}$, and equals no single coefficient in the model. In particular, when $\rho \neq 0$ even “control-only” spillover induces propagation among

treated clusters through the multiplier, so ψ should be read as the structural spillover coefficient on z_i rather than as the marginal indirect effect itself. Accordingly, the simulation study that follows evaluates how well each design recovers the *structural* coefficient β used in the data-generating process, and we are careful throughout not to equate β or ψ with marginal causal contrasts.

D.2 Spatial Dependence

Spatial dependence, used interchangeably with spatial autocorrelation, refers to the property of spatial data where observations close to each other tend not be independent of each other. Rather, observations that are close to each other tend to be dependent on each other and influence values of surrounding data. Hence, there is an implied systematic correlation between data at neighboring locations. Spatial correlation can be positive, where similar values tend to be clustered close together, or negative, where dissimilar values are close together. Appropriate consideration of spatial dependence is necessary for performing valid statistical inference as well as for accurate estimation of effects that are modeled. A pair of common methods for assessing spatial autocorrelation are Moran’s I [40][41] and Geary’s C [42][38]. Moran’s I standardizes the spatial auto-covariance by the variance of the data and Geary’s C utilizes the sum of the squared differences between pairs of data as a measure of co-variation. These test statistics are reliant on the spatial weights matrix. Formal expressions for both statistics are provided in Appendix F.

The spatial weights matrix $\mathbf{W}$ encodes the strength of spatial connections between observations; its elements w_{ij} can be constructed as indicator-based (Rook contiguity) or distance-based weights, both of which are row-normalized to ensure comparability across observations with differing numbers of neighbors. In this study, row-normalized indicator (Rook contiguity) weights are used throughout. The spatial dependence term of the SAR model is therefore

$$\rho \mathbf{W} \mathbf{y} = \rho \sum_{j \neq i} w_{ij} y_{jk'} \quad (13)$$

where j indexes clusters, k' indexes a subject in cluster j , and $w_{ij} = 0$ whenever clusters i and j are not Rook neighbors, so the sum reduces naturally to neighboring clusters only. Full definitions of indicator-based and distance-based weight construction, including row-normalization formulas, are provided in Appendix A.

D.3 Estimation for SAR Model

Maximum likelihood estimation is utilized to gather parameter estimates from the spatial lag model from Section D.1[43][44][45]. The subject-level model to be estimated is

$$y_{ik} = \rho \sum_{j \neq i} w_{ij} y_{jk'} + \alpha + \beta x_i + \psi z_i + \epsilon_{ik} \quad (14)$$

where all terms are as defined in Section D.1. The full likelihood and log-likelihood expressions, along with details of the MLE derivation, are provided in Appendix B. In applied tasks, estimates are obtained via the `lagsarlm()` function from the `spdep` R library[28].

E Simulation Study

A simulation study was carried out to consider the contrast of block stratified and random treatment assignment sampling methods with consideration for spillover effects, spatial dependence, and restrictions on spillover application between intervention and control regions. The results of this study are detailed in the remainder of this section.

E.1 Simulation Scenarios & Response

We carry out a simulation study to explore how estimation error of the intervention effect in a Spatial CRT varies across a range of spillover effect sizes, the presence or absence of spatial dependence, varying restrictions on clusters where spillover is permitted to occur, and the method of sampling for treatment assignment. Each simulation scenario is composed of a set of these parameters. The baseline effect, α , is always assigned as 0.2. The intervention effect, β , is always assigned as 1.0. The spillover effect, ψ , has possible values 0.5, 0.6, 0.7, and 0.8. The spatial dependence parameter, ρ is set as either 0.00 or 0.01. The sampling design for each scenario is either Block Stratified Sampling (BSS) or Simple Random Sampling (SRS). Finally, the spillover application type is (1) Control Clusters Only or (2) Control & Intervention Clusters. Each simulation setting is replicated for each of the spatial grid dimensions discussed previously (2x4, 3x3, and 3x4).

Given that the form of the spatial lag model in Section D.1 contains the response on both sides of the expression, a separate closed-form expression for data generation must be considered[27][46]. This

data generation expression is

$$\mathbf{y} = (\mathbf{I} - \rho \mathbf{W})^{-1} (\boldsymbol{\alpha} + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) \quad (15)$$

where the mean structure is $\alpha + \beta x_i + \psi z_i$, $x_i \in \{0, 1\}$ is the cluster-level indicator of whether cluster i receives the intervention, and $z_i \in \{0, 1\}$ is the cluster-level indicator of whether cluster i receives any spillover effect; both are shared by all subjects k within cluster i .

The estimation model fitted to each simulated dataset is the spatial lag model

$$y_{ik} = \rho \sum_{j \neq i} w_{ij} y_{jk} + \alpha + \beta x_i + \psi z_i + \epsilon_{ik} \quad (16)$$

which mirrors the data-generating structure above and is estimated via maximum likelihood using `lagsarlm()` (Appendix B). The fitted model therefore *includes* the spillover term ψz_i ; bias in $\hat{\beta}$ does not arise from deliberately omitting a variable. Rather, it reflects two sources: (1) collinearity between the treatment indicator x_i and the spillover indicator z_i , a structural feature of the allocation that varies between SRS and BSS; and (2) the interaction of background autocorrelation ρ with the spatial configuration of treatment assignments.

The first source deserves explicit treatment because it is exact, not merely high, for an entire family of designs. Under block stratified sampling with control-only spillover (TrtNoSpill), the checkerboard allocation places every control cluster adjacent to at least one treated cluster while no treated cluster borders another treated cluster; hence the spillover indicator satisfies $z_i = 1 - x_i$ *exactly* for all clusters. The design matrix columns for x_i and z_i are then perfectly collinear, so β and ψ are not separately identifiable. Writing the mean structure as $\alpha + \beta x_i + \psi z_i = (\alpha + \psi) + (\beta - \psi)x_i$ shows that the intercept absorbs ψ and the treatment coefficient is shifted to $\beta - \psi$; consequently $\hat{\beta}$ estimates $\beta - \psi$ and its bias is $-\psi$. This is precisely the pattern observed in the results: for BSS/TrtNoSpill with $\rho = 0$, the estimated bias of $\hat{\beta}$ is -0.496 , -0.597 , -0.697 , and -0.798 at $\psi = 0.5, 0.6, 0.7, 0.8$ respectively (Table 2), matching $-\psi$ to within Monte Carlo error. By contrast, most SRS allocations contain at least one control cluster with no treated Rook neighbor (so $z_i = 0$ there), which breaks the exact collinearity and permits partial separation of β and ψ ; this is the structural reason SRS can recover β more accurately than BSS in the TrtNoSpill regime. Characterizing how these allocation-induced identification challenges propagate into $\hat{\beta}$ across grid geometries, spillover regimes, and sampling designs is the central aim of this simulation study.

BSS is fundamentally a *design-stage* tool: the set of valid BSS allocations is determined prior to data collection, and this study evaluates estimator behavior by exhaustively enumerating all possible treatment allocations for each spatial grid. This exhaustive enumeration is feasible for the small grids studied here (70, 126, and 924 total combinations, respectively) and directly characterizes the full design space. The simulation does not claim to support permutation-based frequentist inference under BSS, nor does it use BSS as a basis for a randomization distribution. Rather, model-based inference via SAR maximum likelihood estimation—specifically the `lagsarlm()` R function—is the appropriate inferential framework for practitioners implementing either SRS or BSS in an applied study. The role of this simulation is to identify which design choice leads to an estimator with more favorable bias-variance properties when the true data-generating model (SAR with spillover) is correctly specified. An important corollary follows from the small number of valid BSS allocations (2 for the 2×4 grid, 6 for the 3×3 grid, and 2 for the 3×4 grid): if a trialist were to adopt BSS as the randomization mechanism, the allocation would be nearly deterministic, severely limiting the power of any permutation-based randomization test. BSS is therefore best understood as a *design-selection criterion*—a principled way to identify a spatially favorable allocation prior to data collection—rather than as a randomization distribution suitable for permutation inference. In applied studies, inference should proceed via model-based SAR maximum likelihood estimation regardless of whether SRS or BSS was used for allocation; this constraint is discussed further in Section 4.2.

E.2 Simulation Procedure

The algorithm implemented to perform the simulations for this study is illustrated in the following section. The simulation is performed over all potential cluster treatment combinations (intervention/control) for each of the 2×4 , 3×3 , and 3×4 cluster spatial grid geometries. The simulation procedure is as follows:

For each spatial grid geometry and simulation scenario, the complete set of possible treatment combinations was enumerated exhaustively (70 for the 2×4 grid, 126 for the 3×3 grid, and 924 for the 3×4 grid). For each combination, $N = 50$ simulation replications were performed for the 2×4 and 3×3 grids and $N = 10$ for the 3×4 grid, with 20 subjects uniformly sampled per cluster in each replication. Outcomes were generated from the SAR closed-form data-generating process (Equation 15), and parameters $(\alpha, \beta, \psi, \rho)$ were estimated via `lagsarlm()` for each replication. BSS-valid combinations were identified

and tracked separately (2 for the 2×4 grid, 6 for 3×3 , and 2 for 3×4). The complete step-by-step algorithm, including BSS identification criteria and aggregation procedures, is provided in Appendix C.

Figure 7 presents an example of the 8 cluster 2×4 spatial grid object. 160 subjects (20 subjects for each cluster) with uniformly distributed coordinates are sampled over the area of the spatial object and colored according to their cluster membership. Subject counts are not necessarily identical between any pair of clusters as the sample of subjects is considered over the whole space. Points in any cluster on the grid are subject to the selected treatment (intervention/control) and equally experience any potential spillover along with all other points that are members of that same cluster. We emphasize that the individual within-cluster coordinates serve only to define cluster membership and to visualize the sampled grid; they do *not* enter the SAR fit. Because x_i , z_i , and the spatial weights w_{ij} are all defined at the cluster level, the estimator depends on a subject only through the cluster it belongs to, and the simulated point locations within a cluster are not used by the model. Similar illustrative figures of the sampled observations over the spatial grid could be constructed and displayed for the other spatial geometries under consideration in the simulation study.

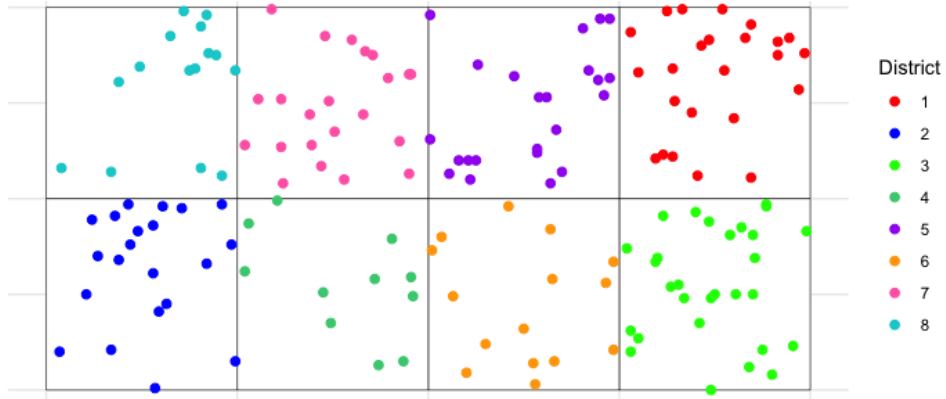


Figure 7: Example of simulated observation coordinates ($n=160$) on a 2×4 grid of clusters

E.3 Simulation Evaluation Metrics

Estimates for the baseline effect (α), intervention effect (β), spillover effect (ψ), and spatial correlation parameter (ρ) were collected for each iteration of the simulation procedure for every treatment assignment combination. Considering the initial fixed simulation parameters, the following metrics are computed for estimation of the intervention effect for the evaluation of each set of simulation conditions.

Bias: a measure of how different the estimated parameter value is from the true parameter value. Small bias means that the parameter estimate is quite close to the true value.

Variance: a measure of the spread of a set of estimates from multiple simulation iterations. Smaller variance indicates that the estimates are tightly grouped together and large variance means that the estimates are spread out.

Mean Squared Error (MSE): measures the average squared difference between each estimate and the true value of a parameter. Mean Squared Error combines variance to understand the consistency of a set of estimates as well as bias to understand any trend in deviation from the true value into a measure of estimation accuracy.

Since $MSE = Bias^2 + Variance$, differences in MSE between SRS and BSS can be attributed to either systematic bias or sampling variability across allocations. This decomposition is visually reflected in the grouped bar figures throughout the Results section (Figures 8–10), where the relative contribution of bias-driven versus variance-driven error is apparent from differences in bar height across sampling methods and conditions.

Mean Squared Error is the primary metric of consideration to understand the efficient estimation of the intervention effect under all simulation settings. The following subsections detail the findings from the simulation study, focusing on the Mean Squared Error (MSE), standard deviation (SD), and bias of the intervention effect (β) estimates under various conditions (simulation scenarios). The analysis compares Simple Random Sampling (SRS) and Block Stratified Sampling (BSS) across different spatial grid configurations, spillover effect magnitudes, spatial dependence levels, and spillover application types.

E.4 Intervention Effect Estimates

We report the Mean Squared Error (MSE), standard deviation (SD), and bias of $\hat{\beta}$ for SRS and BSS across all three grid geometries in Tables 2 and 3, with the corresponding MSE patterns shown in Figures 8–10. Rather than narrate each grid in turn, we organize the findings by spillover regime, since the regime—not the grid size—drives the qualitative contrast between the two designs.

Control-only spillover (TrtNoSpill). Here BSS is generally the stronger design, and on the larger grids the advantage is dramatic: for the 3×3 and 3×4 grids BSS attains near-zero MSE with near-zero SD (e.g., BSS MSE ≈ 0.001 and ≈ 0.0004 at $\psi = 0.5$, $\rho = 0$, versus SRS MSE = 0.092 and 0.034), because the few checkerboard allocations yield reliably consistent estimates. The 2×4 grid is the lone partial exception: BSS still bounds the worst case (its SD is an order of magnitude smaller than SRS throughout), but when $\rho = 0$ its *average* MSE can exceed SRS’s (BSS 0.262 vs. SRS 0.173 at $\psi = 0.5$). This is the exact-collinearity regime analyzed in Section E: with $z_i = 1 - x_i$, $\hat{\beta}$ estimates $\beta - \psi$ and carries a bias of $-\psi$ —the BSS bias entries of Table 2 read -0.496 , -0.597 , -0.697 , -0.798 for the 2×4 grid at $\rho = 0$ —so BSS trades a known, design-induced bias for sharply reduced variance.

Bidirectional spillover (TrtSpill). When spillover also flows among treated clusters, SRS is typically the lower-MSE choice on average, though with substantially larger SD: some random allocations carry considerable estimation risk. The important exception is the *checkerboard paradox* on the 3×4 grid: with spatial dependence present ($\rho = 0.01$), BSS MSE (≈ 0.223 at $\psi = 0.5$) substantially exceeds SRS MSE ($= 0.112$), reversing BSS’s usual advantage. Because the strict checkerboard surrounds every intervention cluster entirely by control neighbors, bidirectional spillover amplifies rather than dampens the systematic bias in $\hat{\beta}$ (discussed further in Section 4).

Common patterns. Across all grids and both regimes, increasing the spillover magnitude ψ raises MSE for both designs, and introducing even slight spatial dependence ($\rho = 0.01$) inflates MSE relative to $\rho = 0$. The overarching message is one of risk management: BSS bounds the worst-case estimation error—near-zero SD across its few allocations—and is most advantageous when spillover is confined to control clusters, whereas SRS can achieve a lower average MSE but exposes the analyst to high-variance “unlucky” allocations and, under bidirectional spillover with spatial dependence, can outperform a BSS design hampered by the checkerboard paradox.

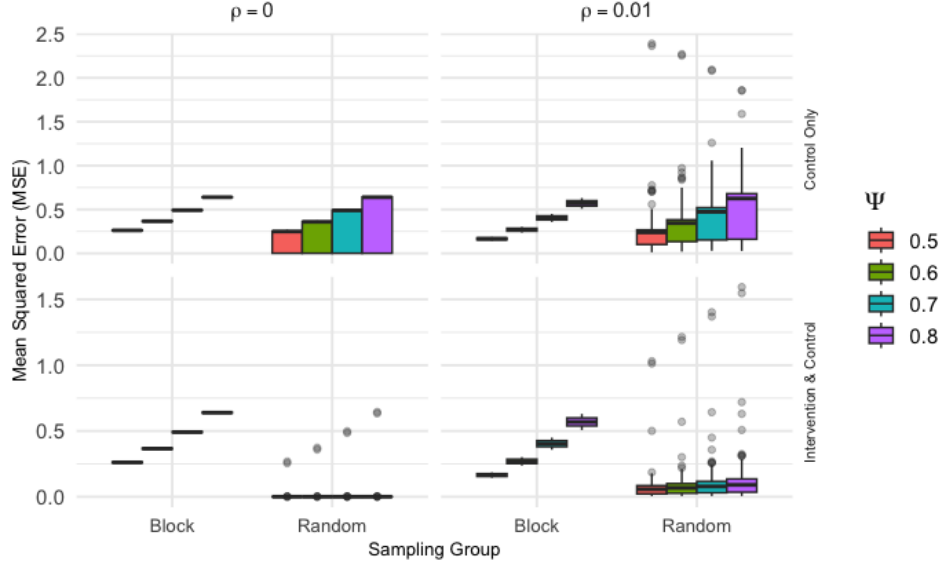


Figure 8: Intervention effect MSE for the 2x4 cluster grid grouped by spillover effect magnitude, spatial dependence, sampling design, and spillover application type.

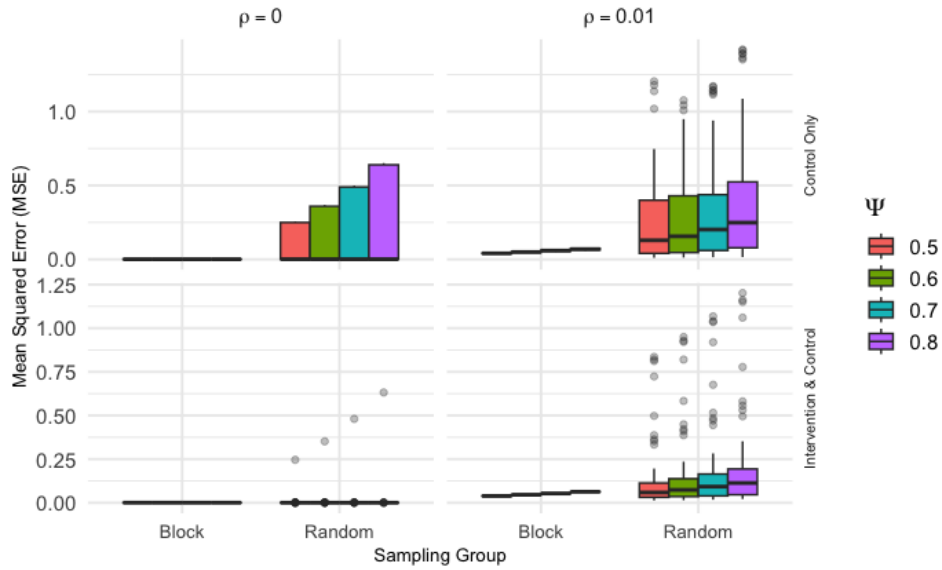


Figure 9: Intervention effect MSE for the 3x3 cluster grid grouped by spillover effect magnitude, spatial dependence, sampling design, and spillover application type.

F Supporting Methods: Spatial-Dependence Diagnostics and Distance-Based Neighbors

Before fitting a spatial model, it is advisable to test for the presence of spatial autocorrelation. Two widely used diagnostics are Moran's I [40][41] and Geary's C [42][38].

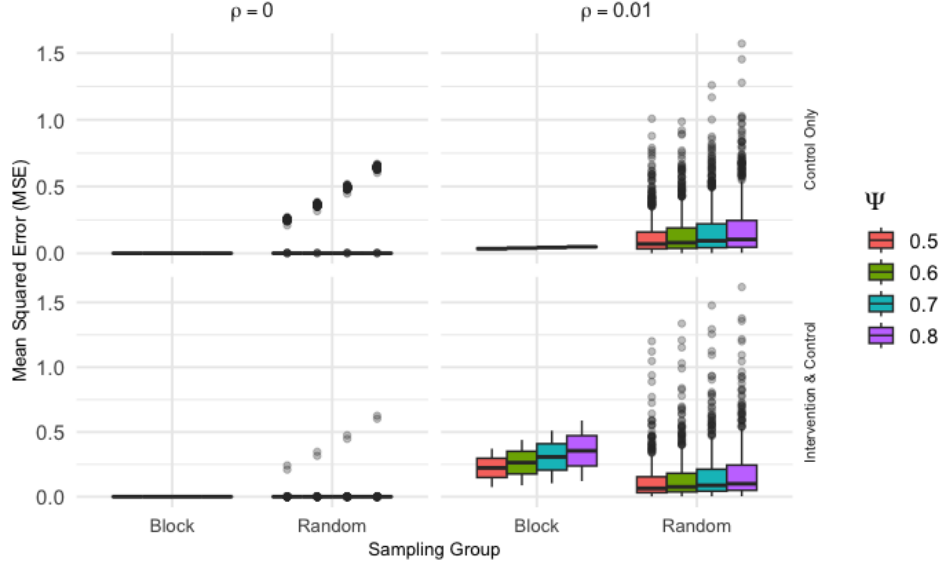


Figure 10: Intervention effect MSE for the 3x4 cluster grid grouped by spillover effect magnitude, spatial dependence, sampling design, and spillover application type.

Moran's I

Moran's I standardizes the spatial auto-covariance by the variance of the data:

$$I = \frac{N \sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\left(\sum_i \sum_j w_{ij} \right) \sum_i (x_i - \bar{x})^2} \quad (17)$$

where N is the number of spatial units, x_i is the value at location i , $\bar{x}$ is the global mean, and w_{ij} are elements of the spatial weights matrix. Values range from -1 (perfect negative autocorrelation) to $+1$ (perfect positive autocorrelation); the expected value under the null hypothesis of no spatial autocorrelation is $E[I] = -(N-1)^{-1}$. Moran's I can be computed in R via `moran.test()` from `spdep`[29].

Geary's C

Geary's C uses the sum of squared differences between adjacent observations:

$$C = \frac{(N-1) \sum_i \sum_j w_{ij} (x_i - x_j)^2}{2 \left(\sum_i \sum_j w_{ij} \right) \sum_i (x_i - \bar{x})^2} \quad (18)$$

where all terms are as defined above. Values significantly below 1 indicate positive spatial autocorrelation; values above 1 indicate negative autocorrelation; and $E[C] = 1$ under the null hypothesis. Geary's C can

Grid	Spatial Dependence (ρ)	Spillover Effect (ψ)	Simple Random Sampling			Block Stratified Sampling		
			MSE	SD	Bias	MSE	SD	Bias
2×4 (SD×10)	Yes (0.01)	0.5	0.304	4.053	-0.270	0.165	0.294	-0.349
		0.6	0.382	4.062	-0.342	0.269	0.462	-0.471
		0.7	0.471	4.143	-0.414	0.404	0.653	-0.594
		0.8	0.573	4.390	-0.486	0.569	0.867	-0.716
	No (0)	0.5	0.173	1.170	-0.343	0.262	0.095	-0.496
		0.6	0.248	1.683	-0.411	0.366	0.100	-0.597
		0.7	0.338	2.291	-0.480	0.492	0.095	-0.697
		0.8	0.441	2.994	-0.548	0.639	0.081	-0.798
3×3 (SD×10)	Yes (0.01)	0.5	0.234	2.593	-0.054	0.041	0.044	-0.012
		0.6	0.276	2.860	-0.092	0.050	0.080	-0.018
		0.7	0.322	3.284	-0.131	0.059	0.114	-0.022
		0.8	0.372	3.857	-0.170	0.069	0.145	-0.025
	No (0)	0.5	0.092	1.207	-0.182	0.001	0.002	0.003
		0.6	0.132	1.740	-0.219	0.001	0.002	0.003
		0.7	0.180	2.370	-0.255	0.001	0.002	0.003
		0.8	0.235	3.099	-0.292	0.001	0.002	0.003
3×4 (SRS SD×100; BSS MSE×10, BSS SD×100)	Yes (0.01)	0.5	0.120	13.244	0.033	0.346	0.963	-0.152
		0.6	0.138	14.807	0.021	0.388	0.505	-0.167
		0.7	0.159	17.166	0.009	0.435	0.011	-0.180
		0.8	0.183	20.387	-0.001	0.484	0.561	-0.193
	No (0)	0.5	0.034	8.554	-0.068	0.004	0.022	-0.006
		0.6	0.049	12.318	-0.081	0.004	0.022	-0.006
		0.7	0.067	16.769	-0.094	0.004	0.020	-0.006
		0.8	0.087	21.912	-0.108	0.003	0.016	-0.006

Table 2: SRS vs. BSS intervention-effect error estimates ($\hat{\beta}$) under *control-only* spillover (TrtNoSpill), for the three grids. Standard deviations (and, for the 3×4 grid, the BSS MSE column) carry the multiplicative factor noted beside each grid; other entries are unscaled. The 2×4 block at $\rho = 0$ shows the exact-collinearity signature, with BSS bias = $-\psi$.

be computed in R via `geary.test()` from `spdep`[29].

Distance-Based Neighbor Construction

Distance-based neighbors[27] are constructed under the assumption that clusters within a specified distance of a reference cluster are considered neighbors. As illustrated in Figure 1 (c), the reference cell (dark) claims as neighbors all cells whose centroids lie within a specified radius $d_{\max}$. Formally, cluster j is a distance-based neighbor of cluster i if $d_{ij} < d_{\max}$, where d_{ij} is the Euclidean distance between the centroids of clusters i and j . Distance-based neighborhoods can also be defined for individual point observations rather than areal units; in that case, any point within $d_{\max}$ of the reference point is a neighbor. This definition generalizes Rook contiguity in the sense that, for a regular grid with appropriately

Grid	Spatial Dependence (ρ)	Spillover Effect (ψ)	Simple Random Sampling			Block Stratified Sampling		
			MSE	SD	Bias	MSE	SD	Bias
2×4 (SD×10)	Yes (0.01)	0.5	0.095	1.762	0.031	0.165	0.294	-0.349
		0.6	0.115	2.082	0.030	0.269	0.462	-0.471
		0.7	0.138	2.423	0.029	0.404	0.653	-0.594
		0.8	0.162	2.787	0.027	0.569	0.867	-0.716
	No (0)	0.5	0.008	0.438	-0.014	0.262	0.095	-0.496
		0.6	0.011	0.614	-0.017	0.366	0.100	-0.597
		0.7	0.015	0.825	-0.020	0.492	0.095	-0.697
		0.8	0.019	1.072	-0.023	0.639	0.081	-0.798
3×3 (SD×10)	Yes (0.01)	0.5	0.103	1.480	0.071	0.039	0.022	-0.019
		0.6	0.122	1.688	0.082	0.046	0.027	-0.020
		0.7	0.143	1.905	0.092	0.054	0.032	-0.021
		0.8	0.168	2.149	0.101	0.063	0.007	-0.019
	No (0)	0.5	0.002	0.219	-0.004	0.001	0.004	0.004
		0.6	0.003	0.313	-0.004	0.001	0.004	0.004
		0.7	0.004	0.428	-0.005	0.001	0.004	0.004
		0.8	0.005	0.562	-0.006	0.001	0.004	0.004
3×4 (SRS SD×100; BSS MSE×10, BSS SD×100)	Yes (0.01)	0.5	0.112	12.873	0.022	2.229	20.995	-0.415
		0.6	0.131	14.762	0.021	2.634	24.750	-0.452
		0.7	0.151	16.817	0.020	3.074	28.774	-0.488
		0.8	0.173	19.046	0.018	3.548	33.060	-0.525
	No (0)	0.5	0.001	1.051	-0.001	0.003	0.004	-0.003
		0.6	0.001	1.542	-0.001	0.003	0.004	-0.002
		0.7	0.001	2.143	-0.002	0.003	0.005	-0.002
		0.8	0.002	2.853	-0.002	0.003	0.005	-0.002

Table 3: SRS vs. BSS intervention-effect error estimates ($\hat{\beta}$) under *bidirectional* spillover (TrtSpill), for the three grids. Scaling factors are as noted beside each grid. The 3×4 grid at $\rho = 0.01$ illustrates the checkerboard paradox, where BSS MSE exceeds SRS.

chosen $d_{\max}$, the distance-based neighbors coincide with the Rook neighbors (or Queen neighbors if $d_{\max}$ is slightly larger). For irregular geographies, distance-based neighbors offer a flexible alternative that does not require shared boundaries. The corresponding distance-based spatial weights and their row normalization are given in Appendix A.